

## Assignment

# Dengue Patient Data Analysis and Interactive Dashboard Development using Power BI

**Aim:** To develop an interactive dengue healthcare analytics dashboard using Microsoft Power BI to analyze patient data, identify positive dengue cases, examine demographic and geographic patterns, and study clinical, fever, temperature, and symptom-related trends through interactive visualizations and data analysis.

### Objectives:

- To analyze dengue patient data and identify the distribution of positive dengue cases and total patients.
- To evaluate patient outcomes based on demographic factors such as gender and area.
- To analyze geographic patterns using a hierarchical structure of District, Area, and HouseType.
- To examine clinical indicators such as platelet count, WBC count, and body temperature.
- To analyze diagnostic test indicators including IgG, IgM, and NSI.
- To study fever duration and symptom-related patterns, including headache.
- To create interactive dashboards using KPIs, slicers, charts, and drill-down analysis in Power BI.
- To derive meaningful healthcare insights from the visualized patient data.

### Tools / Software Required :

- Microsoft Power BI Desktop
- Dengue Patient Dataset

### Theory :

#### Dengue Healthcare Analytics:

Dengue healthcare analytics involves analyzing patient-related information to understand positive cases, patient outcomes, demographic patterns, geographical distribution, clinical indicators, and symptoms. In this experiment, Power BI is used to analyze dengue patient data and present the findings through an interactive dashboard.

#### KPI Analysis:

Key Performance Indicators are numerical measures used to provide a quick summary of important information. The dashboard uses KPIs such as total patients, positive cases, and average platelet count to provide an overview of the dataset.

#### Hierarchical Analysis:

Hierarchical analysis organizes data into multiple levels so that users can move from a broad view to a more detailed view. In this dashboard, the geographic hierarchy is analyzed using District, Area, and HouseType.

#### Interactive Filtering:

Slicers and filters allow users to dynamically select specific categories and examine their effect on the visualizations. Gender and Area are used as interactive filters in the report.

#### Data Visualization:

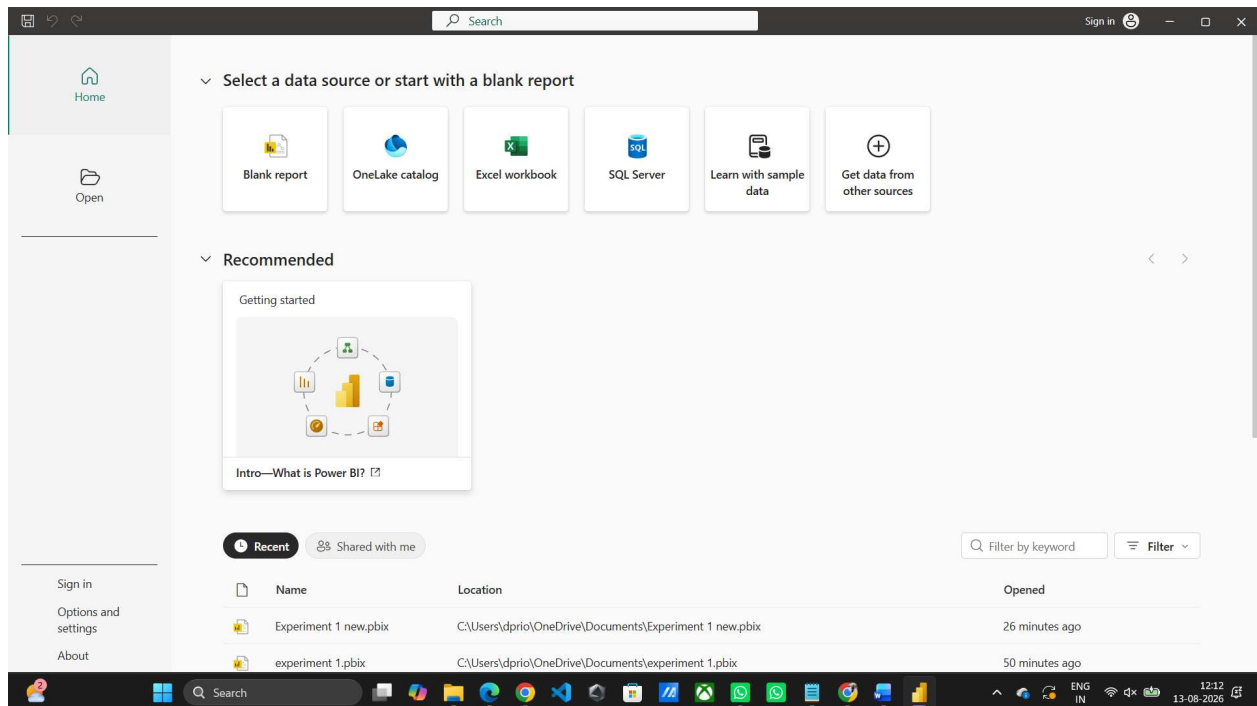
Charts and graphs convert numerical and categorical data into visual information, making relationships and patterns

easier to interpret. The dashboard uses pie charts, bar charts, column charts, KPI cards, scatter plots, and other interactive visuals for analysis.

## Conduct :

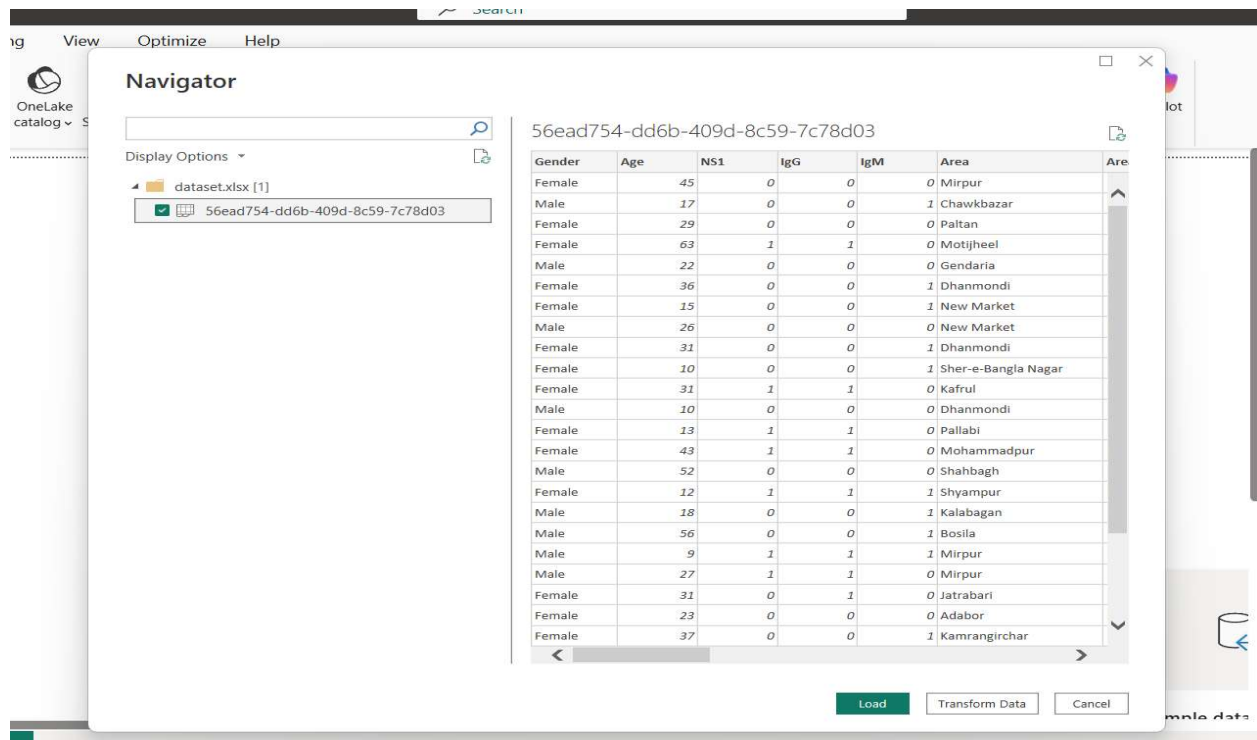
### Part A: Creating and Preparing Tables

**Step 1:** Open **Microsoft Power BI Desktop** and create a new Power BI report.

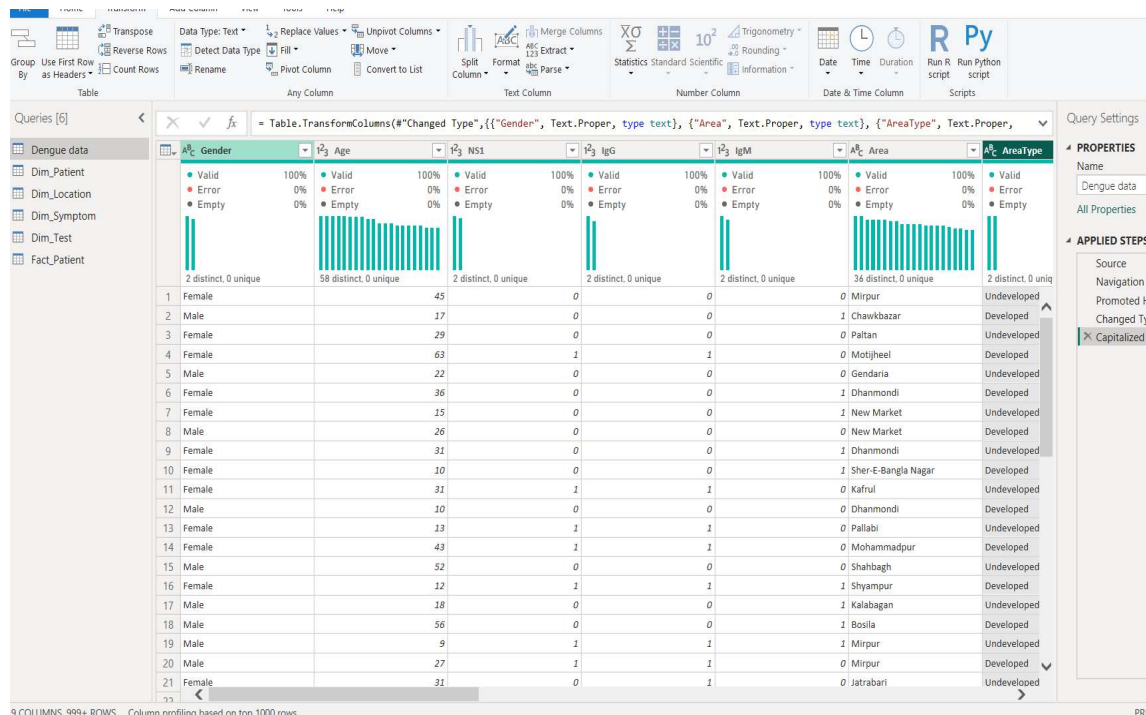


**Step 2:** Import the dengue patient dataset using **Home** → **Get Data** and load the required data into Power BI.

**Step 3:** Open **Power Query Editor** using **Home** → **Transform Data** and inspect the imported data.



**Step 4:** Check the columns and correct their data types. Ensure that numerical fields such as **Platelet\_Count**, **WBC\_Count**, **Body\_Temperature**, **Fever\_Duration**, **IgG**, **IgM** and **NS1** are stored as numeric values, while categorical fields such as **Gender**, **Area**, **District**, **HouseType** and **Outcome** are stored appropriately.



**Step 5:** Create and organize the required tables for the data model:

- **Fact\_Patient** – contains the main patient-related numerical and outcome data.

Table: Fact\_Patient

Columns: Gender, Age, IgM, HouseType, District, Body\_Temperature, Platelet\_Count

Query Settings: Table.TransformColumns(#"Removed Blank Rows",{"HouseType", Text.Proper, type text}, {"District", Text.Proper, type text}, {"Gender", ...})

Properties: Name: Fact\_Patient

Applied Steps: Source, Removed Other Columns, Removed Blank Rows, Capitalized Each Word

| ID | Gender | Age | IgM                   | HouseType  | District | Body_Temperature | Platelet_Count    |
|----|--------|-----|-----------------------|------------|----------|------------------|-------------------|
| 1  | Female | 45  | 2 distinct, 0 unique  | 0 Other    | Dhaka    | 36.9             | 994 distinct, 988 |
| 2  | Male   | 17  | 58 distinct, 0 unique | 0 Other    | Dhaka    | 36.5             |                   |
| 3  | Female | 29  | 2 distinct, 0 unique  | 0 Other    | Dhaka    | 36.6             |                   |
| 4  | Female | 63  | 0 Empty               | 0 Building | Dhaka    | 38.9             |                   |
| 5  | Male   | 22  | 0 Empty               | 1 Other    | Dhaka    | 37               |                   |
| 6  | Female | 36  | 0 Empty               | 1 Building | Dhaka    | 36.9             |                   |
| 7  | Female | 15  | 0 Empty               | 1 Building | Dhaka    | 36.7             |                   |
| 8  | Male   | 26  | 0 Empty               | 0 Other    | Dhaka    | 37               |                   |
| 9  | Female | 31  | 0 Empty               | 1 Tinsed   | Dhaka    | 36.7             |                   |
| 10 | Female | 10  | 0 Empty               | 1 Tinsed   | Dhaka    | 36.8             |                   |
| 11 | Female | 31  | 0 Empty               | 0 Building | Dhaka    | 39.7             |                   |
| 12 | Male   | 10  | 0 Empty               | 0 Tinsed   | Dhaka    | 37.3             |                   |
| 13 | Female | 13  | 0 Empty               | 0 Building | Dhaka    | 38.7             |                   |
| 14 | Female | 43  | 0 Empty               | 0 Building | Dhaka    | 39.9             |                   |
| 15 | Male   | 52  | 0 Empty               | 0 Other    | Dhaka    | 37.1             |                   |
| 16 | Female | 12  | 0 Empty               | 1 Tinsed   | Dhaka    | 39.2             |                   |
| 17 | Male   | 18  | 0 Empty               | 1 Tinsed   | Dhaka    | 37.1             |                   |
| 18 | Male   | 56  | 0 Empty               | 1 Building | Dhaka    | 37.4             |                   |
| 19 | Male   | 9   | 0 Empty               | 1 Tinsed   | Dhaka    | 39.5             |                   |
| 20 | Male   | 27  | 0 Empty               | 0 Tinsed   | Dhaka    | 40.2             |                   |
| 21 | Female | 31  | 0 Empty               | 0 Tinsed   | Dhaka    | 39               |                   |

- **Dim\_Patient** – contains patient-related descriptive information.

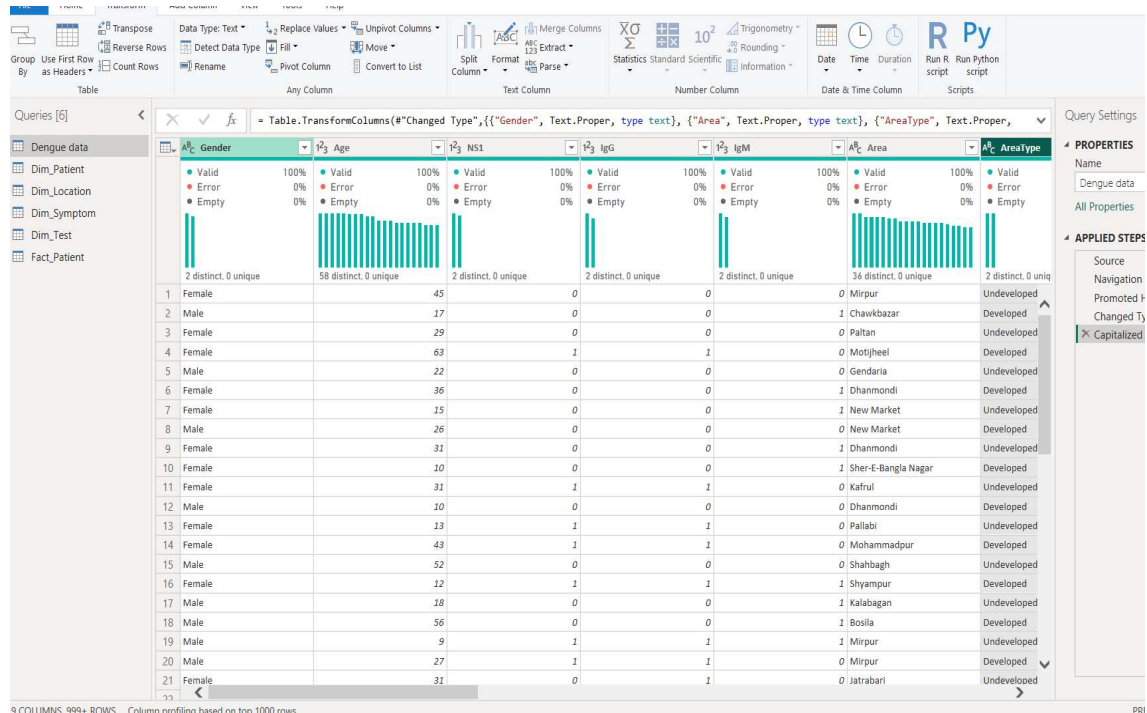
Table: Dim\_Patient

Columns: Gender, Age

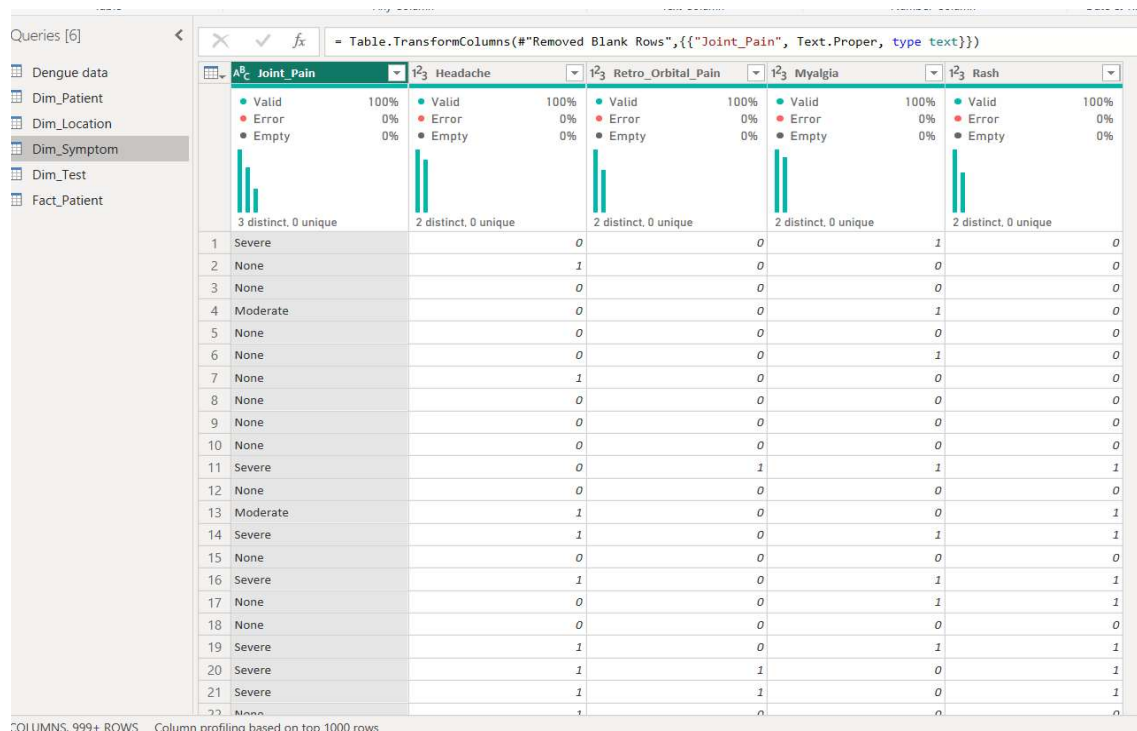
Query Settings: Table.SelectRows(#"Removed Other Columns", each n)

| ID | Gender | Age |
|----|--------|-----|
| 1  | Female | 45  |
| 2  | Male   | 17  |
| 3  | Female | 29  |
| 4  | Female | 63  |
| 5  | Male   | 22  |
| 6  | Female | 36  |
| 7  | Female | 15  |
| 8  | Male   | 26  |
| 9  | Female | 31  |
| 10 | Female | 10  |
| 11 | Female | 31  |
| 12 | Male   | 10  |
| 13 | Female | 13  |
| 14 | Female | 43  |
| 15 | Male   | 52  |
| 16 | Female | 12  |
| 17 | Male   | 18  |
| 18 | Male   | 56  |
| 19 | Male   | 9   |
| 20 | Male   | 27  |
| 21 | Female | 31  |

- **Dim\_Location** – contains geographical information such as District, Area, and HouseType.



- **Dim\_Symptom** – contains symptom-related information such as Headache and Fever Duration.



- **Dim\_Test** – contains diagnostic test information such as IgG, IgM, and NS1.

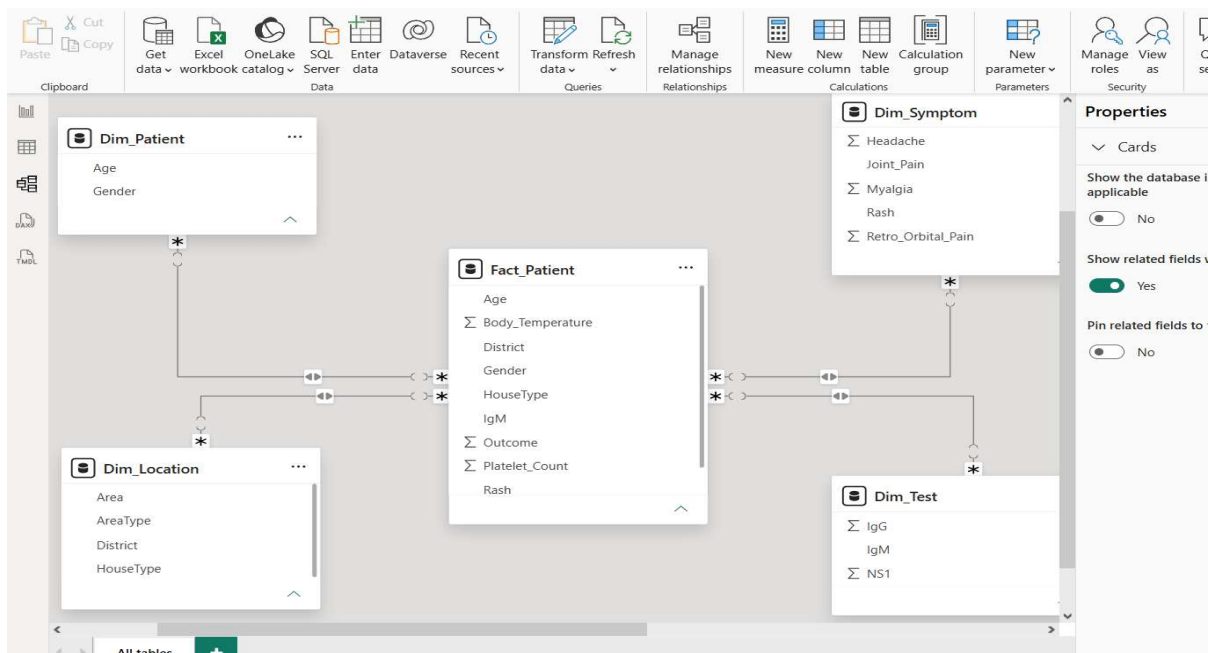
|    | NS1   | IgG   | IgM   |
|----|-------|-------|-------|
| 1  | Valid | Valid | Valid |
| 2  | Valid | Valid | Valid |
| 3  | Valid | Valid | Valid |
| 4  | Valid | Valid | Valid |
| 5  | Valid | Valid | Valid |
| 6  | Valid | Valid | Valid |
| 7  | Valid | Valid | Valid |
| 8  | Valid | Valid | Valid |
| 9  | Valid | Valid | Valid |
| 10 | Valid | Valid | Valid |
| 11 | Valid | Valid | Valid |
| 12 | Valid | Valid | Valid |
| 13 | Valid | Valid | Valid |
| 14 | Valid | Valid | Valid |
| 15 | Valid | Valid | Valid |
| 16 | Valid | Valid | Valid |
| 17 | Valid | Valid | Valid |
| 18 | Valid | Valid | Valid |
| 19 | Valid | Valid | Valid |
| 20 | Valid | Valid | Valid |
| 21 | Valid | Valid | Valid |
| 22 | Valid | Valid | Valid |

**Step 6:** Rename tables and columns where necessary so that they are easy to understand and use while building visualizations.

## Part B: Creating Relationships and Model

**Step 7:** Open **Model View** and arrange the tables so that the relationships can be clearly identified.

**Step 8:** Create the required relationships between the fact table and dimension tables using their common fields or keys.



**Step 9:** Verify the relationship direction and cardinality and ensure that the tables can filter one another correctly.

**Step 10:** Create the required calculated measures for the dashboard, such as:

**1. Total Patients**

Total Patients = COUNTROWS('Dengue data')

**2. Positive Cases**

Positive Cases =

CALCULATE(  
COUNTROWS('Dengue data'),  
'Dengue data'[Outcome] = "Positive"  
)

**3. Average Platelet Count**

Average Platelet Count =

AVERAGE('Dengue data'[Platelet\_Count])

**4. Total Platelet Count**

Total Platelet Count =

SUM('Dengue data'[Platelet\_Count])

**5. Average WBC Count**

Average WBC Count =

AVERAGE('Dengue data'[WBC\_Count])

**6. Average Body Temperature**

Average Body Temperature =

AVERAGE('Dengue data'[Body\_Temperature])

**7. Average Fever Duration**

Average Fever Duration =

AVERAGE('Dengue data'[Fever\_Duration])

## 8. Positive Case Percentage

Positive Case Percentage =

DIVIDE(

[Positive Cases],

[Total Patients],

0

).

## Part C: Creating the Introduction Dashboard

**Step 11:** Open **Report View** and create the Introduction page.

**Step 12:** Add a **Card visual** for Positive Cases.

**Step 13:** Add another Card visual for Total Patients.

**Step 14:** Add a Card visual for Average Platelet Count.

**Step 15:** Create a **Pie Chart** using **Area** and **Positive Cases** to show the distribution of dengue cases across areas.

## Part D: Patient and Positive-Case Overview

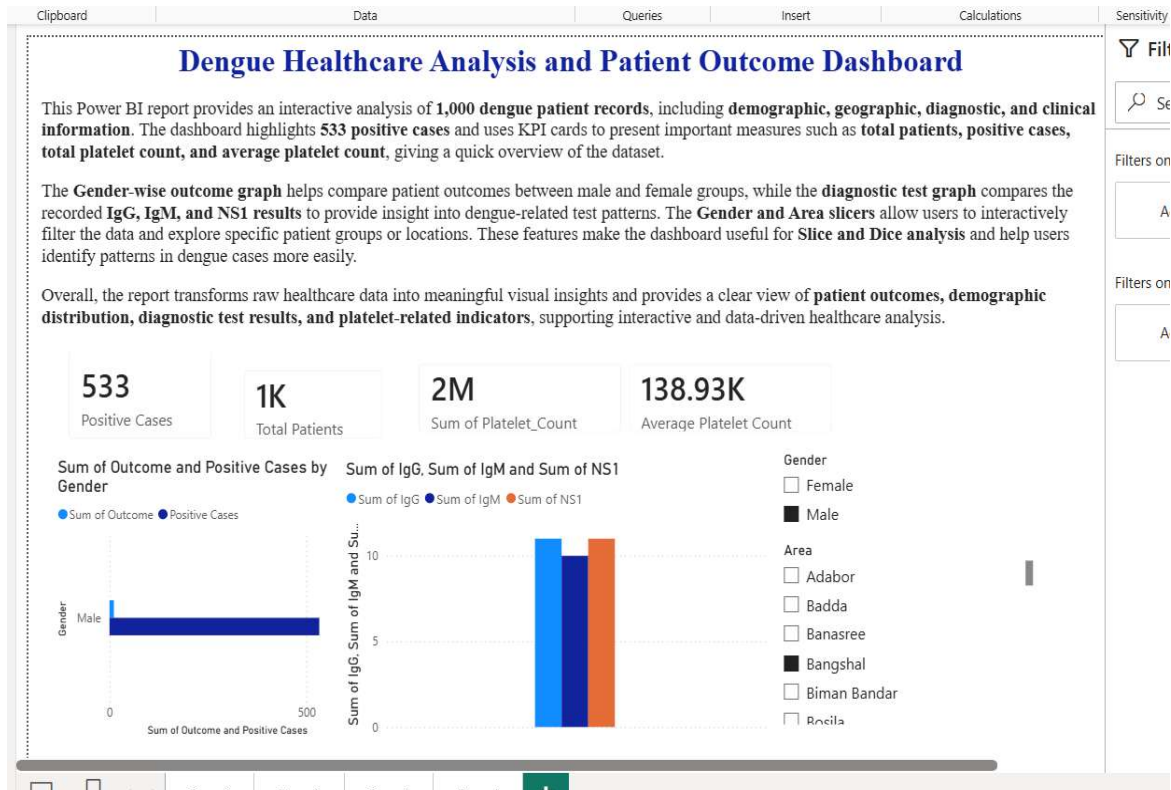
**Step 16:** Create a **Clustered Bar Chart** using Gender and Outcome to compare patient outcomes.

**Step 17:** Add **Gender** and **Area** slicers for interactive filtering.

**Step 18:** Add KPI/Card visuals for Positive Cases and Total Patients.

**Step 19:** Create a clinical measure visual using **Platelet\_Count** and **Average Platelet Count**.

**Step 20:** Create a **Clustered Column Chart** to compare diagnostic indicators **IgG, IgM, and NS1**.



## Part E: Hierarchical Analysis

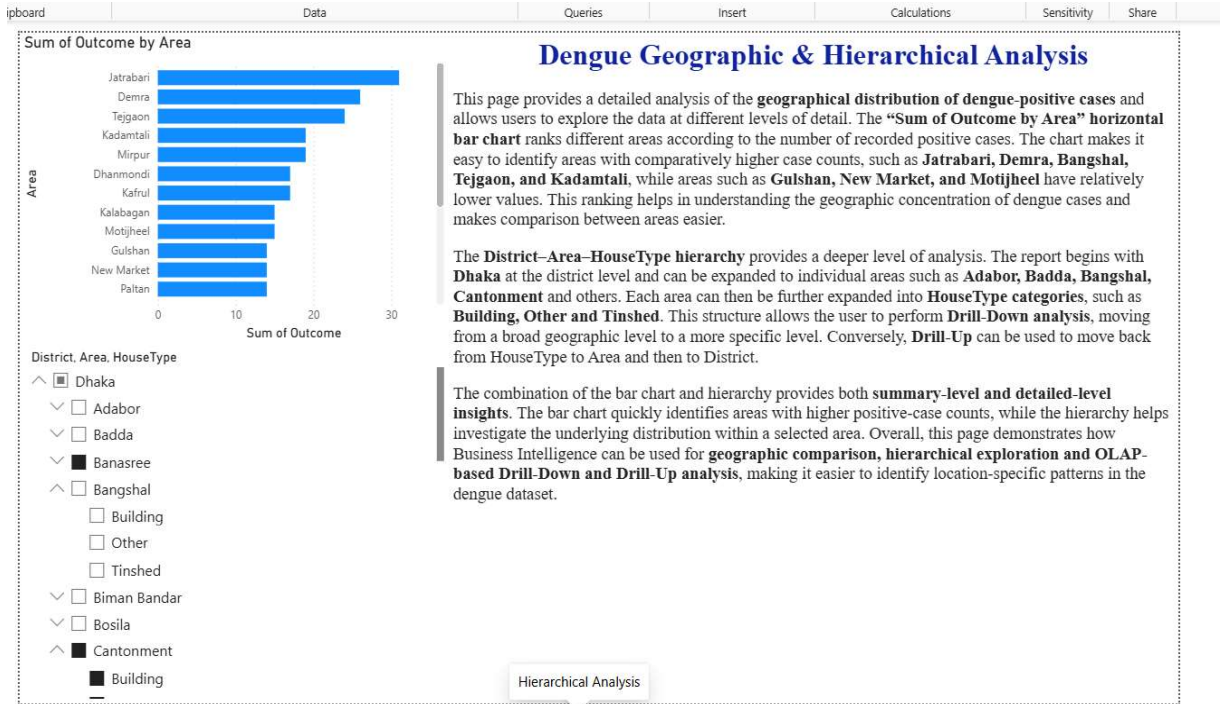
**Step 21:** Create a geographic hierarchy using:

**District → Area → HouseType.**

**Step 22:** Add the hierarchy to the report and create a clustered bar chart for Area-wise Outcome analysis.

**Step 23:** Enable **Drill-Down** so that the user can move from District to Area and then to HouseType.

**Step 24:** Use **Drill-Up** to return to the higher geographic level.



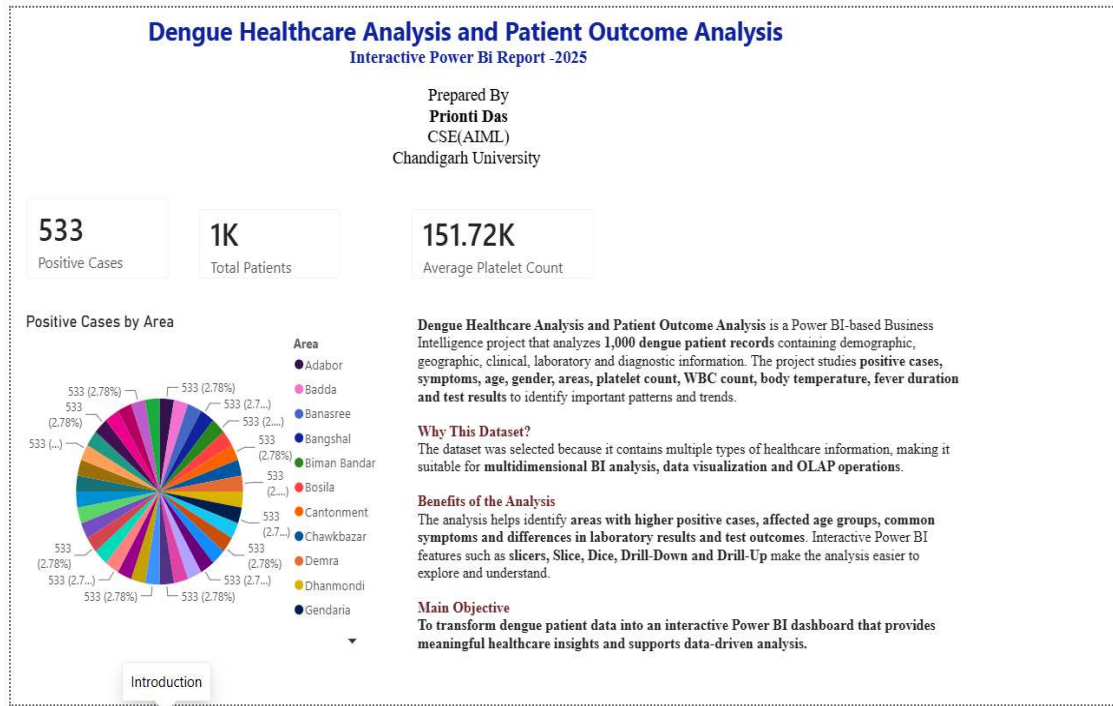
## Part F: Clinical and Patient Outcome Analysis

**Step 25:** Create a **Scatter Chart** using:

- Platelet Count
- WBC Count
- Outcome
- Body Temperature
- Area

**Step 26:** Create a **100% Stacked Bar Chart** to analyze patient outcomes by Gender.

**Step 27:** Create a **Bar Chart** to compare Positive Cases across different Areas.



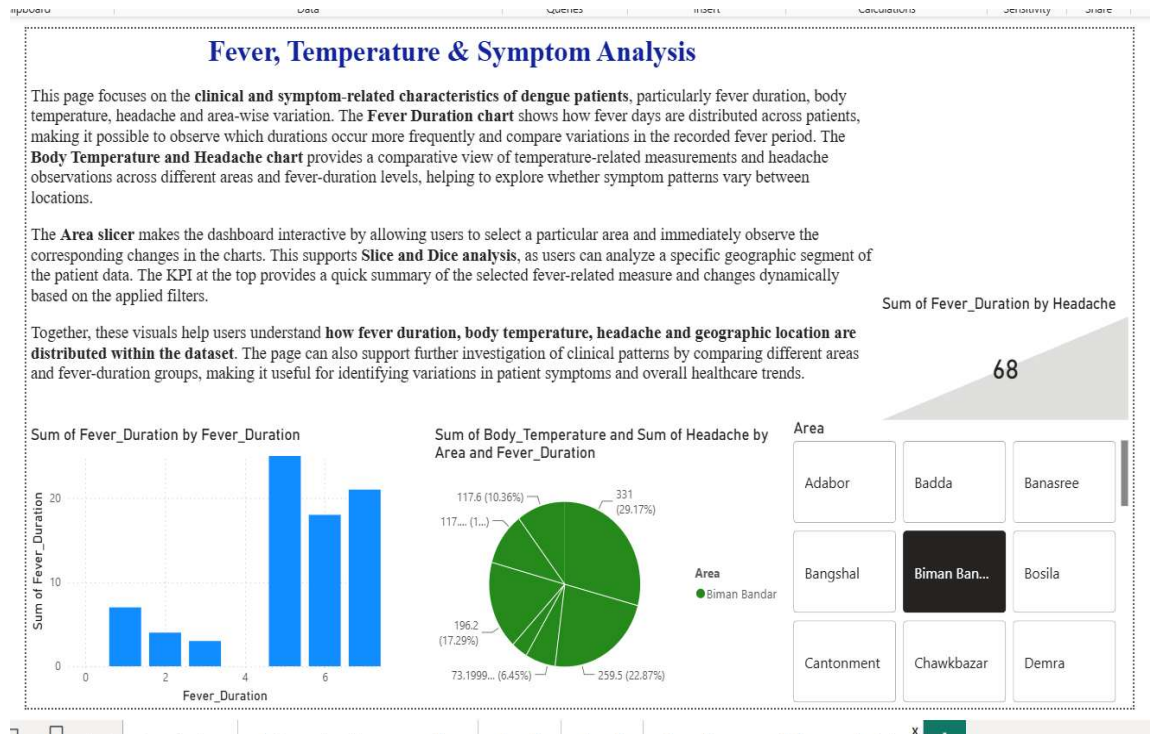
## Part G: Fever, Temperature and Symptom Analysis

**Step 28:** Add an Area slicer for interactive analysis.

**Step 29:** Create a **Column Chart** for Fever Duration.

**Step 30:** Create a visual comparing **Area, Body Temperature, Fever Duration, and Headache**.

**Step 31:** Add a **KPI visual** for fever and symptom-related analysis.

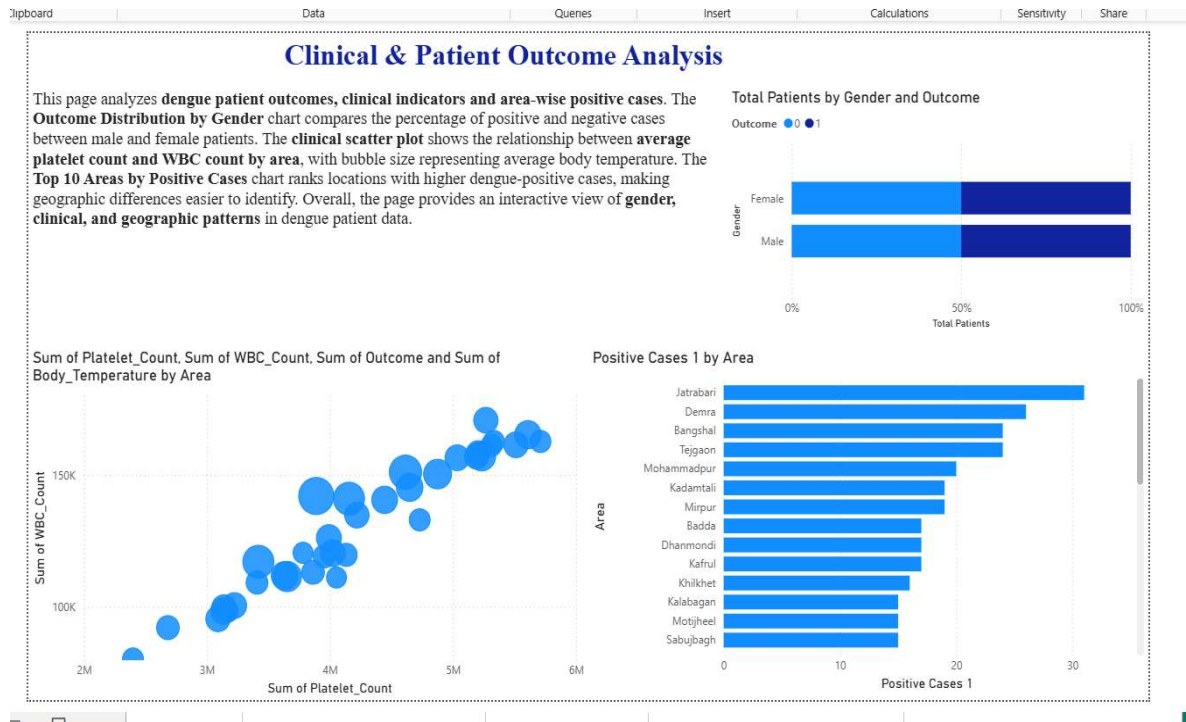


## Part H: Final Analysis

**Step 32:** Test all slicers, filters, and drill-down functions to ensure that the visuals respond correctly.

**Step 33:** Analyze the dashboard to identify patterns in positive dengue cases, patient outcomes, geographic distribution, diagnostic tests, platelet count, WBC count, body temperature, fever duration, and symptoms.

**Step 34:** Record the major insights obtained from the completed Power BI dashboard.



## Result :

The dengue patient dataset was successfully analyzed using Microsoft Power BI. Tables and relationships were created, DAX measures were applied and interactive visualizations were developed to analyze positive cases, patient outcomes, geographic distribution, diagnostic tests, platelet count, WBC count, body temperature, fever duration, and symptoms.

The completed dashboard enables interactive analysis using **KPIs, slicers, filters, hierarchies, Drill-Down and Drill-Up**, helping to identify meaningful patterns and trends in dengue patient data.

## Learning Outcomes :

- Understand the use of Microsoft Power BI for dengue healthcare data analysis.
- Create and organize tables and relationships for a Power BI data model.
- Create DAX measures for calculating important healthcare KPIs.
- Design interactive visualizations using charts, KPI cards, slicers, and filters.
- Perform hierarchical analysis using **District** → **Area** → **HouseType** with Drill-Down and Drill-Up.
- Analyze patient outcomes, positive cases, clinical indicators, fever, temperature, diagnostic tests, and symptoms.
- Interpret the visualized data and derive meaningful healthcare insights.